But now, when Slim and Jim calculate the ratio of the circumference of the ride to its radius they will get a number that is larger than our answer of two times pi, since the circumference is longer but the radius is the same. This is weird. How in the world can something in the shape of a circle violate the ancient Greek realization that for any circle this ratio is exactly two times pi?

Here is Einstein's explanation. The ancient Greek result holds true for circles drawn on a flat surface. But just as the warped or curved mirrors in an amusement park fun-house distort the normal spatial relationships of your reflection, if a circle is drawn on a warped or curved surface, its usual spatial relationships will also be distorted: the ratio of its circumference to its radius will generally not be two times pi.

For instance, Figure 3.2 compares three circles whose radii are identical. Notice, however, that their circumferences are not the same. The circumference of the circle in (b), drawn on the curved surface of a sphere, is less than the circumference of the circle drawn on the flat surface in (a), even though they have the same radius. The curved nature of the sphere's surface causes the radial lines of the circle to converge toward each other slightly, resulting in a small decrease in the circle's circumference. The circumference of the circle in (c), again drawn on a curved surface—a saddle shape—is greater than that drawn on a flat surface; the curved nature of the saddle's surface causes the radial lines of the circle to splay outward from each other slightly, resulting in a small increase in the circle's circumference. These observations imply that the ratio of the circumference to the radius of the circle in (b) will be less than two times pi, while the same ratio in (c) will be greater than two times pi. But this deviation from two times pi, especially the larger value found in (c), is just what we found for the spinning Tornado ride. This led Einstein to propose an idea—the curving of space—as an explanation for the violation of "ordinary," Euclidean geometry. The flat geometry of the Greeks, taught to schoolchildren for thousands of years, simply does not apply to someone on the spinning ride. Rather, its curved space generalization as schematically drawn in part (c) of Figure 3.4 takes its place.5

And so Einstein realized that the familiar geometrical spatial relationships codified by the Greeks, relationships that pertain to "flat"